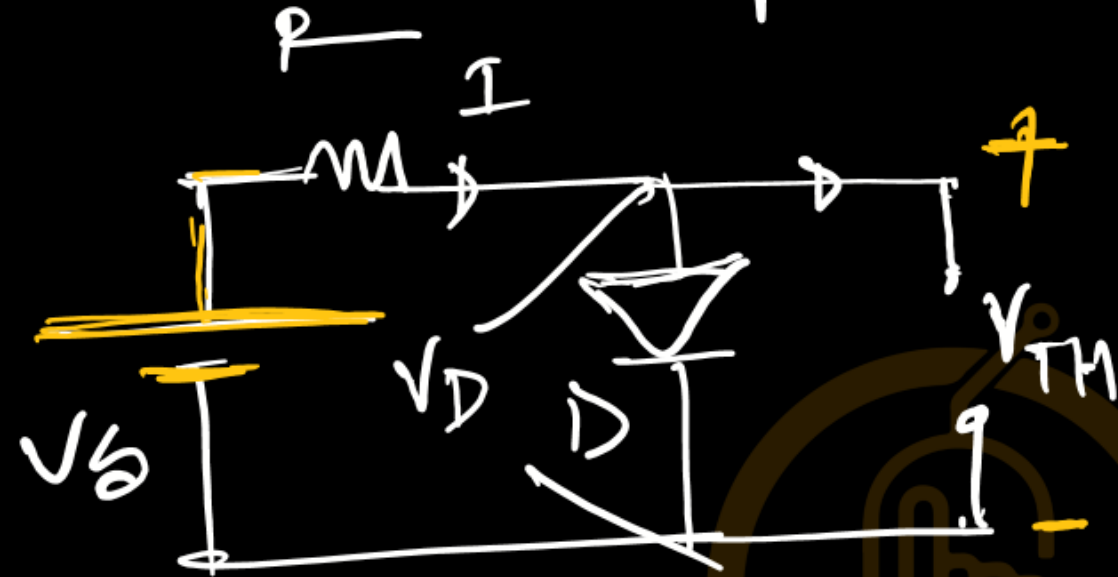


JOB
↓
Room Study.



Analog Electronics

*



Diode
Zener

condⁿ

Diode active

(i) $V_{TH} \geq V_D \rightarrow 0.7 \text{ ভোল্ট} > 0.7 \text{ V}$

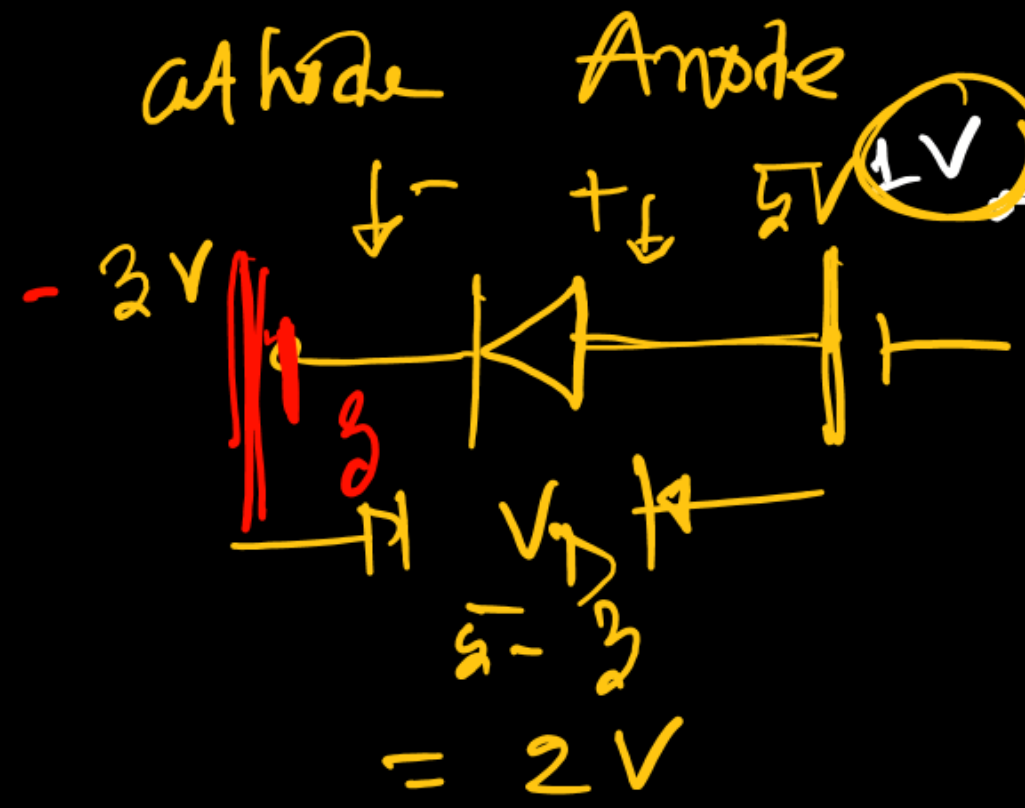
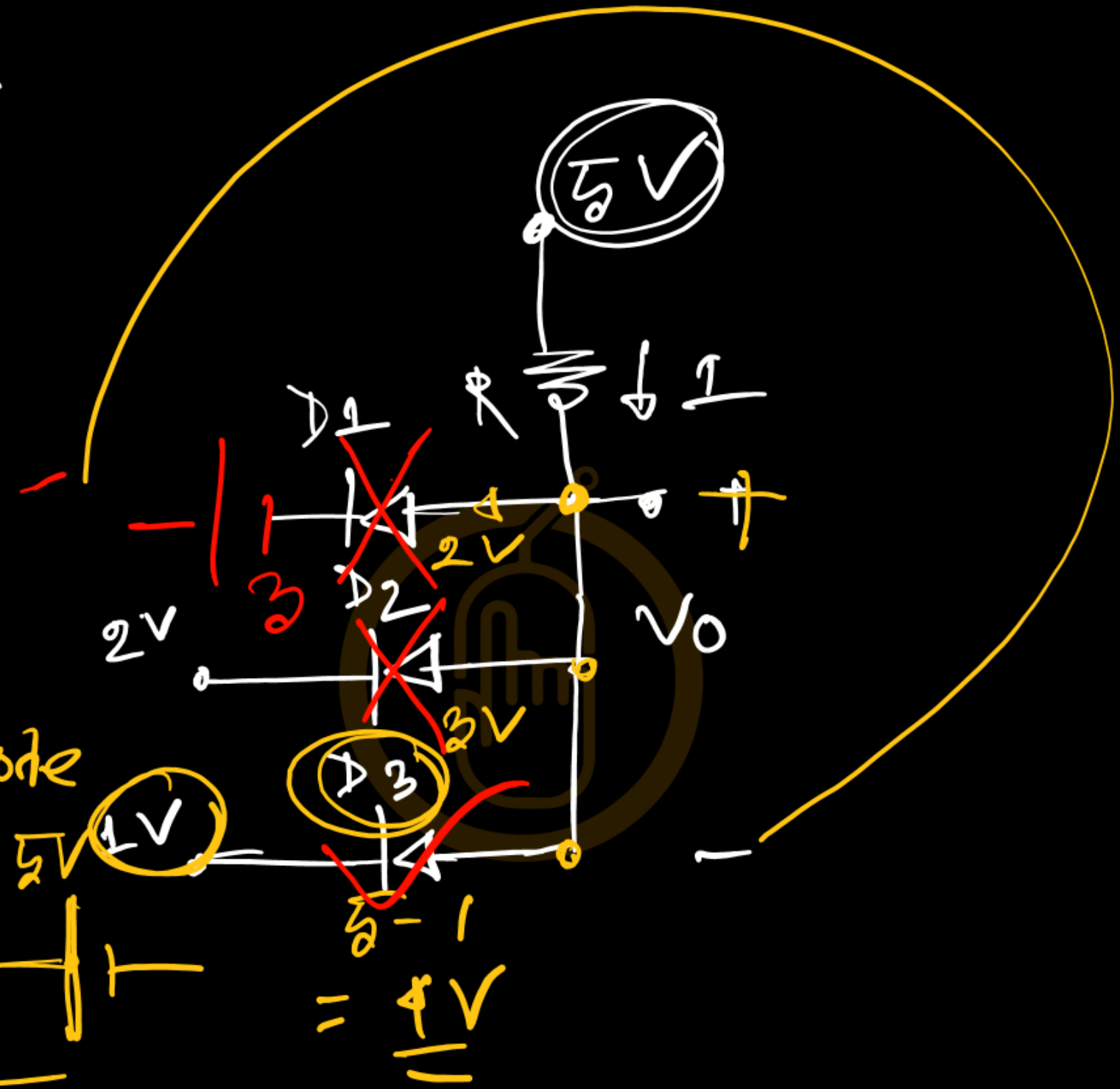
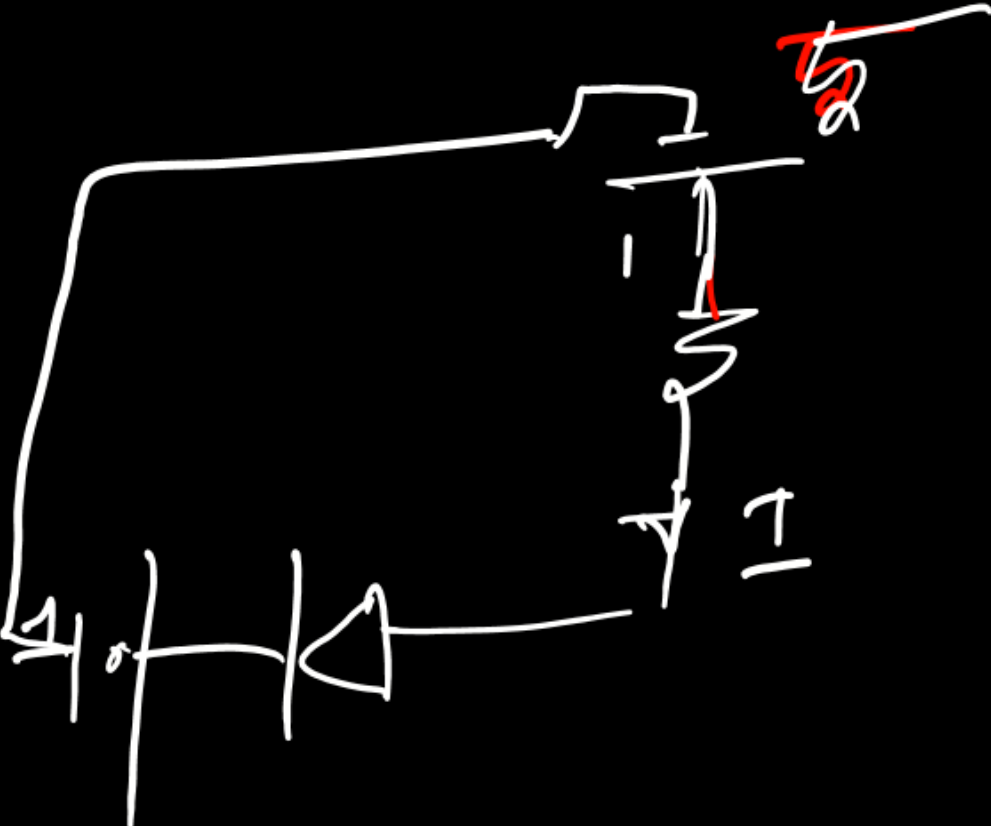
(ii) $V_{TH} < V_D \rightarrow \text{Diode off.}$

$V_i = 0.7 \text{ V}$
 $V_e = 0.3 \text{ V}$

In ideal diode
 $V = 0 \text{ V}$



দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...



$$5 - V_D + 2 = 0$$
$$\Rightarrow 5 + 2 = V_D$$
$$V_D = 7V$$

Zener

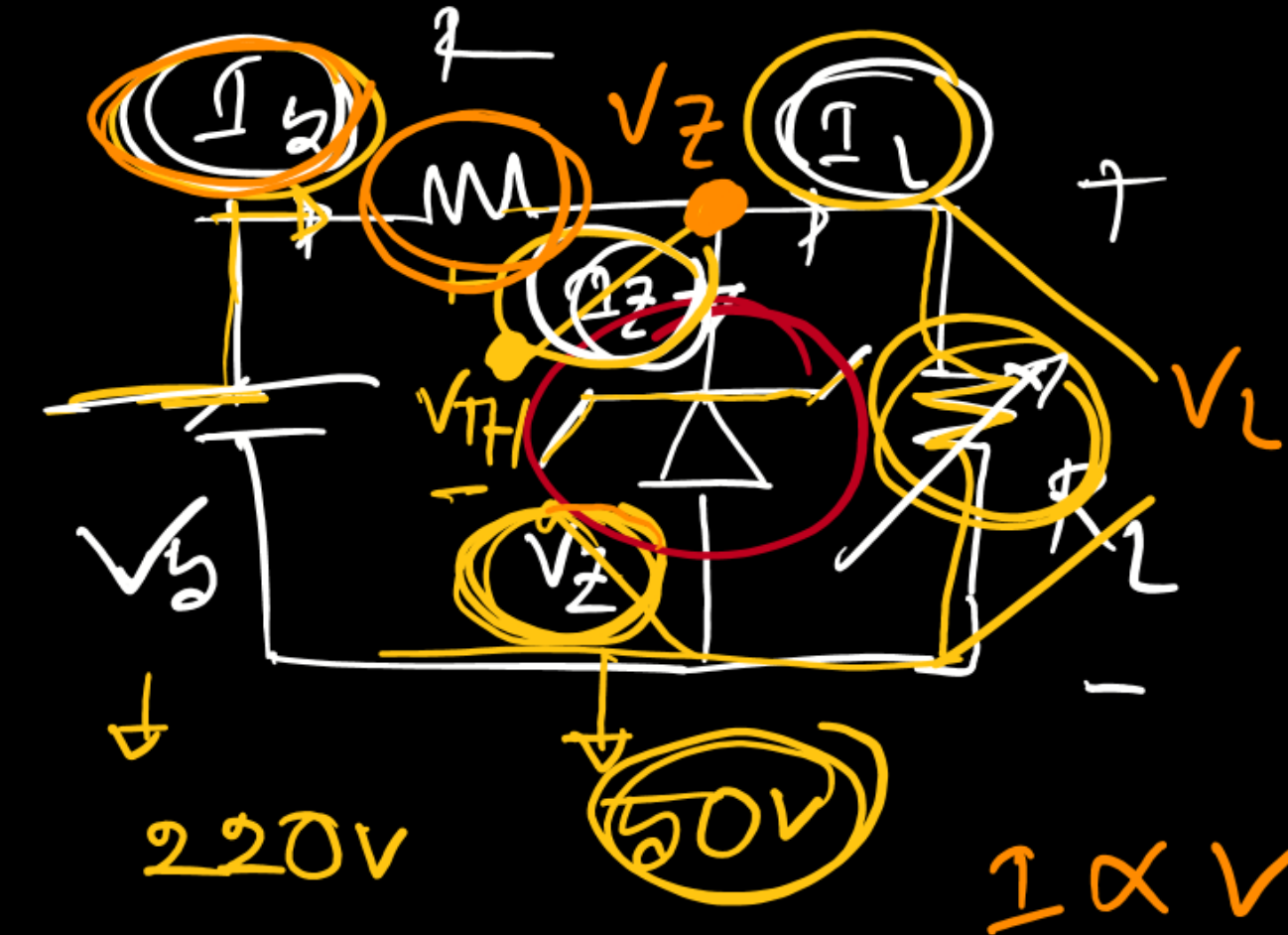
$\eta = 0.01$
 $\eta = 4.95\%$

I_{min}

(i) $V_{TH} \geq V_Z$

(ii) $V_{TH} < V_Z$

$V_Z = V_L$



(i) Load variable $\rightarrow$

$I_Z = I_Z + I_L$
 $I_L \rightarrow 0$
 $I_Z(max) \Rightarrow \frac{V_S - V_Z}{R_{min}} = 0 +$
 $R_{min} =$

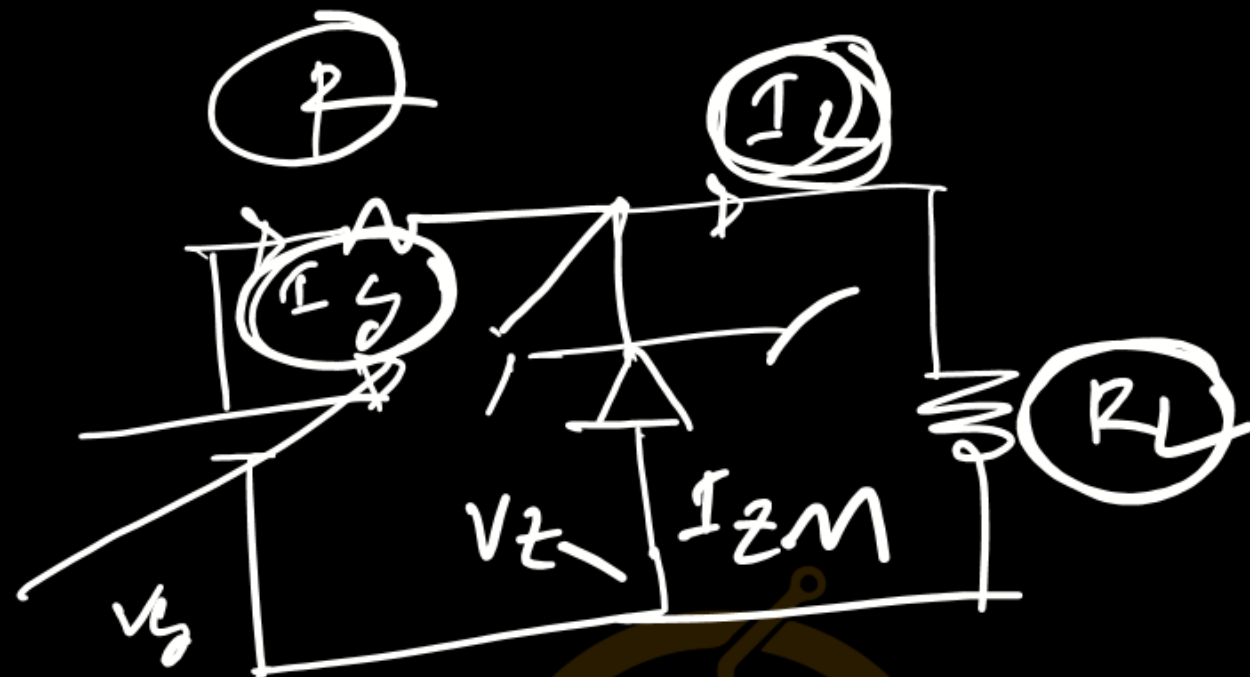
I_{Lmax}
 I_L

$V_L = V_Z$
 R_{min}

$I = \frac{V}{R_{min}}$
 $I \propto \frac{1}{R}$

$V_S > V_Z$
 $50V = 50V$
 $50V < 50$

$R = 10$



$$I_L = \frac{V_z}{R_L}$$

$$V \propto I$$

50 ~ 70

$V_{in(max)} \rightarrow I_{L(max)} \rightarrow I_{zm(max)}$

$V_{in(min)} \rightarrow I_{L(min)} \rightarrow I_{zm(min)}$

$R_L = \text{constant} \rightarrow I_L = \text{fixed}$

Rated value is always Max^m Value.



$V_{in\ max}$
 $V_{in\ (min)}$

$$\begin{aligned} & \downarrow \\ & I_G = \cancel{I_Z} + \cancel{I_R} \\ \Rightarrow & \frac{V_{i\ max} - V_Z}{R} = \cancel{I_{Z\ max}} + \frac{V}{R_L} \end{aligned}$$

$$\begin{aligned} \Rightarrow & V_{i(max)} = () \\ & \frac{V_{in\ min} - V_b}{R} = I_{Z\ min} + \frac{V}{R_L} = () \end{aligned}$$

$$I_L(\max) = \frac{V_i}{R_{L\min}} \quad [V_z = V_L]$$

$$R_{L\min} = ()$$

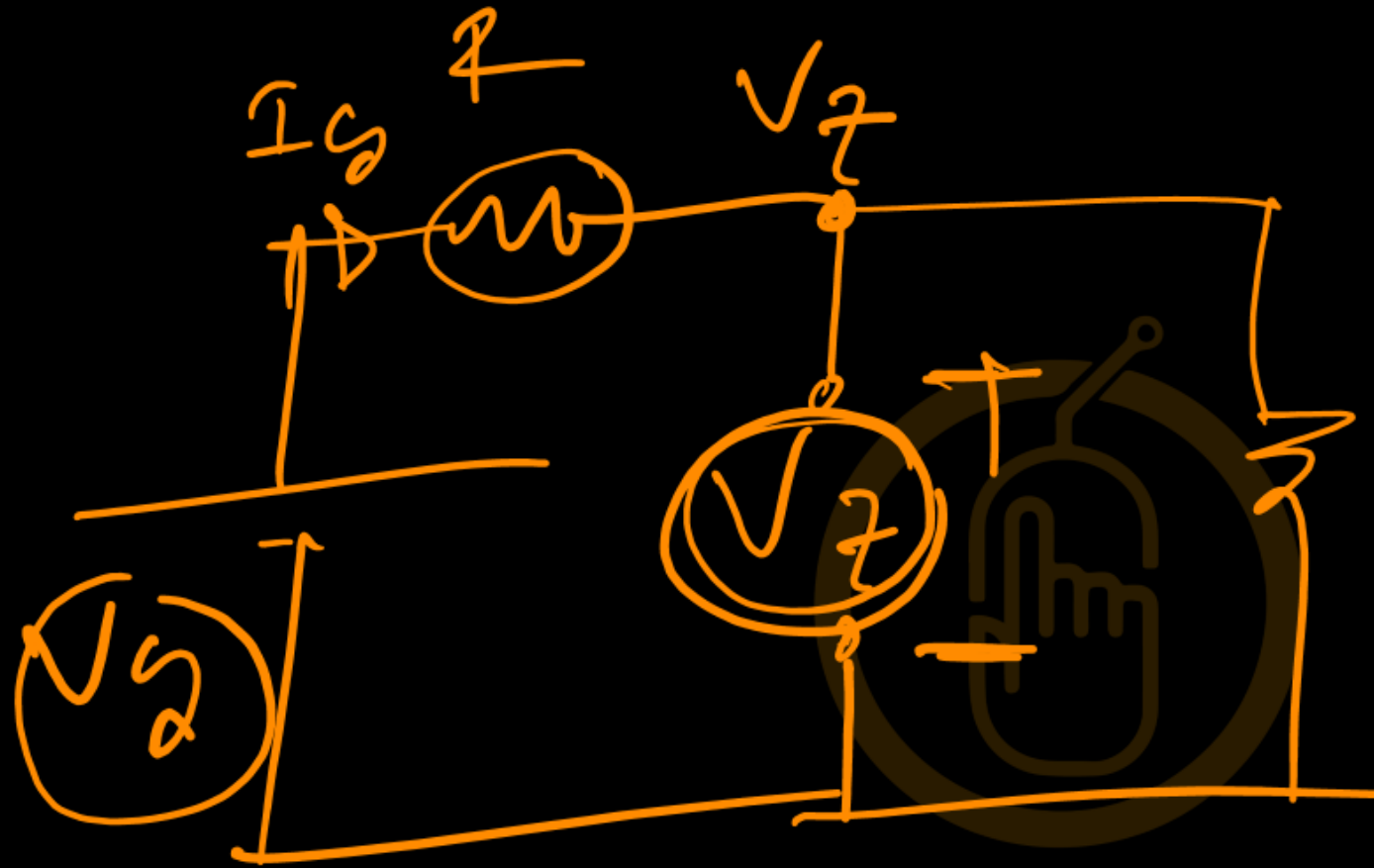
$$I_L(\min)$$

$$I_z = (I_z) + I_L$$

$$I_{zM} = 25 \text{ mA}$$

$$\Rightarrow \frac{V_z - V_t}{R} = 25 + I_L$$

$$I_{L\min} = \frac{V}{R_{L\max}} \rightarrow R_{L(\max)} = ()$$



$$V_2 - I_2 R - V_z = 0$$

$$\Rightarrow V_2 - V_z = I_2 R$$

$$I_2 = \frac{V_2 - V_z}{R}$$



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চাকরি যুদ্ধে...